AQA A-Level Further Mathematics (7367) — Topic 1.6

1.6: Further Vectors

1. Scalar (Dot) Product — Review

Dot product: For vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$$

where θ is the angle between them. In component form (3D):

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Vectors are **perpendicular** iff $\mathbf{a} \cdot \mathbf{b} = 0$.

2. Vector (Cross) Product

Cross product: For vectors $\mathbf{a}$ and $\mathbf{b}$ in 3D:

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin\theta \, \hat{\mathbf{n}}$$

where $\hat{\mathbf{n}}$ is the unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ (right-hand rule).

Component formula:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}$$

Properties of the Cross Product

- $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$ — anti-commutative
- $\mathbf{a} \times \mathbf{a} = \mathbf{0}$
- $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ iff $\mathbf{a}$ and $\mathbf{b}$ are parallel
- $|\mathbf{i} \times \mathbf{j}| = |\mathbf{j} \times \mathbf{k}| = |\mathbf{k} \times \mathbf{i}| = 1$; $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$
- Distributive: $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$

Example: Find $\mathbf{a} \times \mathbf{b}$ where $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix}$$
$$= (2 \cdot 6 - 3 \cdot 5)\mathbf{i} - (1 \cdot 6 - 3 \cdot 4)\mathbf{j} + (1 \cdot 5 - 2 \cdot 4)\mathbf{k}$$
$$= (12 - 15)\mathbf{i} - (6 - 12)\mathbf{j} + (5 - 8)\mathbf{k} = -3\mathbf{i} + 6\mathbf{j} - 3\mathbf{k} = \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$$

Area Using the Cross Product

$$\text{Area of parallelogram} = |\mathbf{a} \times \mathbf{b}|$$
$$\text{Area of triangle} = \frac{1}{2}|\mathbf{a} \times \mathbf{b}|$$

3. Scalar Triple Product

Scalar triple product:

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

The **volume of a parallelepiped** with edges $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ is $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$.

Vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are **coplanar** iff $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$.

Example: Find the volume of the parallelepiped with edges $\mathbf{a} = \mathbf{i} + 2\mathbf{j}$, $\mathbf{b} = 3\mathbf{j} + \mathbf{k}$, $\mathbf{c} = \mathbf{i} + \mathbf{j} + \mathbf{k}$.

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 1(3-1) - 2(0-1) + 0 = 2 + 2 = 4$$

Volume $= |4| = 4$.

4. Equations of Lines in 3D

Vector equation of a line:

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$$

where $\mathbf{a}$ is a position vector of a point on the line and $\mathbf{d}$ is the direction vector.

Cartesian (symmetric) form:

$$\frac{x - a_1}{d_1} = \frac{y - a_2}{d_2} = \frac{z - a_3}{d_3}$$

Angle Between Two Lines

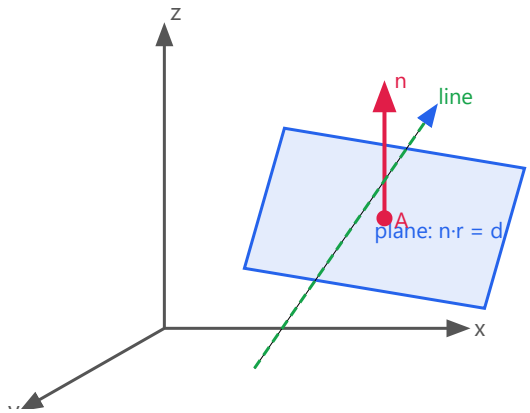
The acute angle θ between lines with directions $\mathbf{d}_1$ and $\mathbf{d}_2$:

$$\cos\theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|}$$

5. Equations of Planes

Vector equation of a plane: $\mathbf{r} \cdot \mathbf{n} = d$, where $\mathbf{n}$ is the normal vector and $d = \mathbf{a} \cdot \mathbf{n}$ for any point $\mathbf{a}$ on the plane.

Cartesian equation: $n_1 x + n_2 y + n_3 z = d$



Angle Between a Line and a Plane

The angle ϕ between a line with direction $\mathbf{d}$ and a plane with normal $\mathbf{n}$:

$$\sin \phi = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$$

(Note: the angle between a line and a plane is the complement of the angle between the line and the normal.)

Angle Between Two Planes

The angle between two planes equals the angle between their normals $\mathbf{n}_1$ and $\mathbf{n}_2$:

$$\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}$$

6. Distance Formulae

Distance from a Point to a Plane

$$d = \frac{|n_1x_0 + n_2y_0 + n_3z_0 - D|}{|\mathbf{n}|}$$

where (x_0, y_0, z_0) is the point and $n_1x + n_2y + n_3z = D$ is the plane.

Distance Between Two Parallel Lines

Let $\ell_1 : \mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$ and $\ell_2 : \mathbf{r} = \mathbf{b} + \mu\mathbf{d}$. The distance is:

$$d = \frac{|(\mathbf{b} - \mathbf{a}) \times \mathbf{d}|}{|\mathbf{d}|}$$

Shortest Distance Between Two Skew Lines

For skew lines $\ell_1 : \mathbf{r} = \mathbf{a} + \lambda\mathbf{d}_1$ and $\ell_2 : \mathbf{r} = \mathbf{b} + \mu\mathbf{d}_2$:

$$d = \frac{|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$$

Example: Find the shortest distance between the skew lines $\ell_1 : \mathbf{r} = \begin{pmatrix}1\\0\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\1\end{pmatrix}$ and $\ell_2 : \mathbf{r} = \begin{pmatrix}0\\1\\0\end{pmatrix} + \mu\begin{pmatrix}0\\1\\1\end{pmatrix}$.

$$\mathbf{d}_1 \times \mathbf{d}_2 = \begin{pmatrix}1\\0\\1\end{pmatrix} \times \begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}0-1\\0-1\\1-0\end{pmatrix} = \begin{pmatrix}-1\\-1\\1\end{pmatrix}$$

$$|\mathbf{d}_1 \times \mathbf{d}_2| = \sqrt{3}$$

$$\mathbf{b} - \mathbf{a} = \begin{pmatrix}-1\\1\\0\end{pmatrix}$$

$$(\mathbf{b}-\mathbf{a})\cdot\begin{pmatrix}-1\\-1\\1\end{pmatrix}=1-1+0=0$$

Shortest distance $=0/\sqrt{3}=0$. (Lines actually intersect — check if skew by testing for solution to $\ell_1=\ell_2$.)

7. Intersections

Line and Plane

Substitute the line equation $\mathbf{r}=\mathbf{a}+\lambda\mathbf{d}$ into the plane equation $\mathbf{r}\cdot\mathbf{n}=D$ and solve for λ.

Two Planes

Two non-parallel planes intersect in a line. Find the direction as $\mathbf{n}_1\times\mathbf{n}_2$; find a point by setting one variable to a convenient value and solving the system.

Three Planes — Analysis

A system of 3 planes can have:

- **Unique intersection point:** $\det(\text{coefficient matrix})\neq 0$
- **A line of intersection:** Two planes intersect in a line that lies on the third
- **No intersection (inconsistent):** Various configurations — triangular prism, or all parallel, etc.

AQA Exam Tip: For 3-plane systems, AQA often gives a parameter k in one equation and asks for the value(s) of k that give different types of intersection. Use $\det A=0$ for the singular case, then test consistency.

Example: Find the point of intersection of the line $\mathbf{r}=\begin{pmatrix}1\\2\\-1\end{pmatrix}+\lambda\begin{pmatrix}1\\-1\\2\end{pmatrix}$ and the plane $2x+y-z=3$.

Substitute: $2(1+\lambda)+(2-\lambda)-(-1+2\lambda)=3$

$2 + 2\lambda + 2 - \lambda + 1 - 2\lambda = 3$
$5 - \lambda = 3 \Rightarrow \lambda = 2$

Point: $\begin{pmatrix} 1+2 \\ 2-2 \\ -1+4 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}$

8. Summary of Key Results

Result	Formula
Dot product	$\mathbf{a} \cdot \mathbf{b} = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \cos\theta$
Cross product magnitude	$\lvert\mathbf{a} \times \mathbf{b}\rvert = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \sin\theta$
Area of parallelogram	$\lvert\mathbf{a} \times \mathbf{b}\rvert$
Volume of parallelepiped	$\lvert\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})\rvert$
Equation of line	$\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$
Equation of plane	$\mathbf{r} \cdot \mathbf{n} = d$
Point to plane distance	$\lvert n_1 x_0 + n_2 y_0 + n_3 z_0 - d\rvert / \lvert\mathbf{n}\rvert$
Skew lines distance	$\lvert(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)\rvert / \lvert\mathbf{d}_1 \times \mathbf{d}_2\rvert$

9. Practice Checklist

- Can you compute dot products and cross products in component form?
- Can you find the area of a triangle given three vertices using the cross product?
- Can you find and use the scalar triple product?
- Can you find angles between lines, between a line and a plane, and between two planes?
- Can you find the distance from a point to a plane?

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- Can you find the intersection of a line with a plane?
- Can you analyse when a 3-plane system has no solution, a line of solutions, or a unique solution?

Vector Product — Full Properties and Applications

For vectors $\mathbf{a}$ and $\mathbf{b}$, the vector (cross) product $\mathbf{a} \times \mathbf{b}$ is a vector perpendicular to both, with magnitude $|\mathbf{a}||\mathbf{b}| \sin \theta$.

Properties:

- $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ (anti-commutative)
- $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$ (distributive)
- $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ (parallel vectors)
- $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$

Area applications:

- Area of parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$: $|\mathbf{a} \times \mathbf{b}|$
- Area of triangle with sides $\mathbf{a}$ and $\mathbf{b}$: $\frac{1}{2}|\mathbf{a} \times \mathbf{b}|$
- Volume of parallelepiped: $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$
- Volume of tetrahedron $OABC$: $\frac{1}{6}|\overrightarrow{OA} \cdot (\overrightarrow{OB} \times \overrightarrow{OC})|$

Example: Find the area of triangle OAB with $O = (0, 0, 0)$, $A = (2, 1, 0)$, $B = (1, 3, 1)$. $\mathbf{a} = (2, 1, 0)$, $\mathbf{b} = (1, 3, 1)$.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{vmatrix} = (1 - 0)\mathbf{i} - (2 - 0)\mathbf{j} + (6 - 1)\mathbf{k} = (1, -2, 5)$$

Area $= \frac{1}{2}\sqrt{1 + 4 + 25} = \frac{\sqrt{30}}{2}$.

Lines and Planes — Distance Calculations

Distance from a point to a plane

The perpendicular distance from point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz = d$ is:

$$\text{dist} = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Distance between skew lines

For skew lines $\mathbf{r}_1 = \mathbf{a}_1 + s\mathbf{d}_1$ and $\mathbf{r}_2 = \mathbf{a}_2 + t\mathbf{d}_2$:

$$\text{dist} = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$$

Example: Find the distance from $P(1, 2, 3)$ to the plane $2x - y + 2z = 5$.

$$d = \frac{|2(1) - 1(2) + 2(3) - 5|}{\sqrt{4 + 1 + 4}} = \frac{|2 - 2 + 6 - 5|}{3} = \frac{1}{3}$$

Distance from a point to a line

For point P and line through A with direction $\hat{\mathbf{d}}$ (unit vector):

$$\text{dist} = |\overrightarrow{AP} - (\overrightarrow{AP} \cdot \hat{\mathbf{d}})\hat{\mathbf{d}}| = \frac{|\overrightarrow{AP} \times \mathbf{d}|}{|\mathbf{d}|}$$

Three Planes — Cases of Consistency

Consider three planes Π_1, Π_2, Π_3. The system of equations may have:

Case	Geometric picture	Algebraic test
Unique solution	Three planes meeting at a single point	$\det(A) \neq 0$ for the coefficient matrix
Line of intersection	Three planes sharing a common line (sheaf)	$\det(A) = 0$, system consistent, one free variable
No solution — triangular prism	Three planes forming a prism (no common point)	$\det(A) = 0$, system inconsistent

No solution — parallel planes	Two (or more) planes parallel but not coincident	Parallel normals, different constants

Determining the type (row reduction method): Given: $\Pi_1 : x + y + z = 6$, $\Pi_2 : 2x + y - z = 3$, $\Pi_3 : x - y + 2z = 4$. Augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & -1 & 3 \\ 1 & -1 & 2 & 4 \end{array}\right)$$

$R_2 \to R_2 - 2R_1$: $(0, -1, -3 | -9)$. $R_3 \to R_3 - R_1$: $(0, -2, 1 | -2)$. $R_3 \to R_3 - 2R_2$: $(0, 0, 7 | 16)$. $\det \neq 0$ (rank 3) → unique solution. From $7z = 16$: $z = 16/7$, etc.

AQA Exam Technique — Further Vectors

Common errors and how to avoid them:

- **Cross product sign:** Use the determinant formula carefully. A common error is the sign on the $\mathbf{j}$ component — it has a minus: $-\det\begin{pmatrix} a_1 & a_3 \\ b_1 & b_3 \end{pmatrix}$.
- **Angle between planes:** Use $\cos\theta = \dfrac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}$ (absolute value to get acute angle). Angle between a line and plane uses $\sin\alpha = \dfrac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$.
- **Skew lines:** To prove skew, show: (1) not parallel (direction vectors not scalar multiples), and (2) the system from setting them equal is inconsistent.
- **Foot of perpendicular:** Parameterise the line, substitute into the plane equation to find t, then evaluate the point.
- **Scalar triple product:** $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$. Zero means coplanar.

Vector Equation of a Plane — Forms and Conversions

Three equivalent forms of a plane with normal $\mathbf{n} = (a, b, c)$:

1. **Vector form:** $\mathbf{r} \cdot \mathbf{n} = d$
2. **Cartesian form:** $ax + by + cz = d$
3. **Parametric form:** $\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c}$ where $\mathbf{b}, \mathbf{c}$ lie in the plane

To convert parametric → Cartesian: find $\mathbf{n} = \mathbf{b} \times \mathbf{c}$, then $d = \mathbf{a} \cdot \mathbf{n}$.

Example — Convert to Cartesian:
$\mathbf{r} = (1, 0, 2) + s(1, 2, 0) + t(0, 1, 3)$
$\mathbf{n} = (1, 2, 0) \times (0, 1, 3) = (6 - 0, 0 - 3, 1 - 0) = (6, -3, 1)$
$d = (1, 0, 2) \cdot (6, -3, 1) = 6 + 0 + 2 = 8$
Cartesian: $6x - 3y + z = 8$

Intersection of Lines and Planes — Full Worked Examples

Example 1 — Line meets plane:
Line: $\mathbf{r} = (1, 2, 0) + t(1, -1, 2)$. Plane: $2x + y - z = 3$.
Sub: $2(1 + t) + (2 - t) - (2t) = 3 \Rightarrow 2 + 2t + 2 - t - 2t = 3 \Rightarrow 4 - t = 3 \Rightarrow t = 1$.
Point: $(2, 1, 2)$.

Example 2 — Two planes intersect in a line:
Plane 1: $x + y - z = 2$. Plane 2: $2x - y + z = 1$.
Add: $3x = 3 \Rightarrow x = 1$. Sub: $1 + y - z = 2 \Rightarrow y - z = 1$. Let $z = t$: $y = 1 + t$.
Line: $\mathbf{r} = (1, 1, 0) + t(0, 1, 1)$.

Example 3 — Angle between line and plane:
Line direction: $\mathbf{d} = (1, 2, 2)$. Plane normal: $\mathbf{n} = (2, 1, -2)$.
$$\sin\alpha = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|} = \frac{|2 + 2 - 4|}{3 \cdot 3} = \frac{0}{9} = 0.$$
So $\alpha = 0^\circ$: the line is parallel to the plane (or lies in it). Check: sub a point on the line into the plane equation to distinguish.

Skew Lines — Distance Formula

Distance between skew lines $\mathbf{r} = \mathbf{a} + s\mathbf{b}$ and $\mathbf{r} = \mathbf{c} + t\mathbf{d}$:

$$D = \frac{|(\mathbf{c} - \mathbf{a}) \cdot (\mathbf{b} \times \mathbf{d})|}{|\mathbf{b} \times \mathbf{d}|}$$

Example: Lines $\ell_1 : \mathbf{r} = (0,0,0) + s(1,2,3)$ and $\ell_2 : \mathbf{r} = (1,0,0) + t(1,1,0)$.
$\mathbf{b} \times \mathbf{d} = (1,2,3) \times (1,1,0) = (0-3, 3-0, 1-2) = (-3,3,-1)$. $|\mathbf{b} \times \mathbf{d}| = \sqrt{9+9+1} = \sqrt{19}$.
$\mathbf{c} - \mathbf{a} = (1,0,0)$. $(1,0,0) \cdot (-3,3,-1) = -3$.
$D = \dfrac{3}{\sqrt{19}}$

Vector Product — Scalar Triple Product Applications

The **scalar triple product** $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ gives the signed volume of the parallelepiped with edges $\mathbf{a}, \mathbf{b}, \mathbf{c}$.
Volume of tetrahedron $OABC = \dfrac{1}{6}|[\mathbf{a}, \mathbf{b}, \mathbf{c}]|$.

Example — Volume of tetrahedron: $A = (1,0,0),\ B = (0,2,0),\ C = (0,0,3)$, O origin.
$\mathbf{a} = (1,0,0),\ \mathbf{b} = (0,2,0),\ \mathbf{c} = (0,0,3)$.
$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = 6$.
Volume of tetrahedron $= \dfrac{1}{6} \cdot 6 = 1$.

Coplanarity test: Points A, B, C, D are coplanar iff $\overrightarrow{AB} \cdot (\overrightarrow{AC} \times \overrightarrow{AD}) = 0$.

Reflection of a Point in a Plane

To reflect point P in plane Π:

1. Find foot of perpendicular F from P to Π: parameterise the normal through P as $\mathbf{r} = \mathbf{p} + t\mathbf{n}$, sub into plane equation to find t.
2. Reflection $P' = 2F - P$.

Example: Reflect $P = (3, 1, 0)$ in the plane $x + 2y - z = 5$.
Normal: $\mathbf{n} = (1, 2, -1)$. Line through P: $(3 + t, 1 + 2t, -t)$.
Sub into plane: $(3 + t) + 2(1 + 2t) - (-t) = 5 \Rightarrow 5 + 6t = 5 \Rightarrow t = 0$.
So P already lies in the plane! Reflection is P itself.

Example 2: Reflect $P = (1, 1, 1)$ in the plane $x + y + z = 0$.
Normal $\mathbf{n} = (1, 1, 1)$. Line: $(1 + t, 1 + t, 1 + t)$. Sub: $3 + 3t = 0 \Rightarrow t = -1$. $F = (0, 0, 0)$.
$P' = 2(0, 0, 0) - (1, 1, 1) = (-1, -1, -1)$.

Closest Point on a Line to a Given Point

Given line $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ and external point P:
The foot of perpendicular F has parameter $t = \dfrac{(\mathbf{p} - \mathbf{a}) \cdot \mathbf{b}}{|\mathbf{b}|^2}$.
Distance from P to line: $d = |\mathbf{p} - \mathbf{a} - t\mathbf{b}|$, or $d = \dfrac{|(\mathbf{p} - \mathbf{a}) \times \mathbf{b}|}{|\mathbf{b}|}$.

Example: $P = (4, 0, 0)$, line $\mathbf{r} = (0, 0, 0) + t(1, 1, 1)$.
$t = \dfrac{(4, 0, 0) \cdot (1, 1, 1)}{3} = \dfrac{4}{3}$. $F = \left(\frac{4}{3}, \frac{4}{3}, \frac{4}{3}\right)$.
$\overrightarrow{FP} = \left(\frac{8}{3}, -\frac{4}{3}, -\frac{4}{3}\right)$. Distance $= \sqrt{\frac{64}{9} + \frac{16}{9} + \frac{16}{9}} = \sqrt{\frac{96}{9}} = \dfrac{4\sqrt{6}}{3}$.

Cross Product — Full Properties and Applications

For vectors $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1)$$

Properties:

- $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ (anti-commutative)
- $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$ where θ is the angle between them
- $\mathbf{a} \times \mathbf{a} = \mathbf{0}$
- Area of parallelogram with sides $\mathbf{a}, \mathbf{b}$: $|\mathbf{a} \times \mathbf{b}|$
- Area of triangle OAB: $\frac{1}{2}|\mathbf{a} \times \mathbf{b}|$

Example — Area of triangle: $A = (1, 0, 0),\ B = (0, 2, 0),\ C = (0, 0, 3)$.
$\overrightarrow{AB} = (-1, 2, 0),\ \overrightarrow{AC} = (-1, 0, 3)$.
$\overrightarrow{AB} \times \overrightarrow{AC} = (6 - 0, 0 - (-3), 0 - (-2)) = (6, 3, 2)$.
Area of $\triangle ABC = \frac{1}{2}|(6, 3, 2)| = \frac{1}{2}\sqrt{36 + 9 + 4} = \dfrac{7}{2}$.

Angles Between Lines and Between Planes

Angle between two lines with direction vectors $\mathbf{d}_1, \mathbf{d}_2$:

$$\cos\theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|} \quad \text{(take acute angle)}$$

Angle between two planes with normals $\mathbf{n}_1, \mathbf{n}_2$:

$$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}$$

Example: Find the angle between planes $x + y + z = 1$ and $2x - y + z = 3$.
$\mathbf{n}_1 = (1, 1, 1),\ \mathbf{n}_2 = (2, -1, 1)$.

$$\cos\theta = \frac{|2-1+1|}{\sqrt{3}\cdot\sqrt{6}} = \frac{2}{\sqrt{18}} = \frac{2}{3\sqrt{2}} = \frac{\sqrt{2}}{3}.$$

$$\theta = \arccos\left(\frac{\sqrt{2}}{3}\right) \approx 61.9^\circ.$$

Parametric Lines — Worked Problems

Example 1 — Intersection of two lines:
$\ell_1 : \mathbf{r} = (1,2,3) + s(1,0,1)$ and $\ell_2 : \mathbf{r} = (2,3,4) + t(0,1,1)$.
From components: $1+s=2 \Rightarrow s=1$; $2=3+t \Rightarrow t=-1$; $3+s=4+t \Rightarrow 4=3$.
Inconsistent — lines are **skew**.

Example 2 — Lines that intersect:
$\ell_1 : \mathbf{r} = (0,0,0) + s(1,1,0)$ and $\ell_2 : \mathbf{r} = (1,0,1) + t(0,1,-1)$.
$s=1,\ s=t,\ 0=1-t \Rightarrow t=1,\ s=1$. Check all three: consistent ✓. Intersection: $(1,1,0)$.

Section Formula and Ratio Division

Point dividing AB in ratio $m:n$ internally: $\mathbf{r} = \dfrac{n\mathbf{a}+m\mathbf{b}}{m+n}$.
Midpoint ($m:n=1:1$): $\mathbf{r} = \dfrac{\mathbf{a}+\mathbf{b}}{2}$.

Example: $A=(1,2,3),\ B=(4,5,6)$. Point P divides AB in ratio $2:1$.
$$\mathbf{p} = \frac{1\cdot(1,2,3)+2\cdot(4,5,6)}{3} = \frac{(9,12,15)}{3} = (3,4,5).$$

Vector Identities and Triple Products

Vector triple product: $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$
Lagrange's identity: $|\mathbf{a}\times\mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a}\cdot\mathbf{b})^2$

Further Vectors AQA Exam Technique:

- Always convert between forms of plane equation smoothly — show the cross product computation clearly.
- For skew lines proof: show non-parallel (directions not scalar multiples) *then* show inconsistency of simultaneous equations.
- Distance formulas: don't confuse point-to-line ($\frac{|(\mathbf{p}-\mathbf{a})\times\mathbf{b}|}{|\mathbf{b}|}$) and point-to-plane ($\frac{|\mathbf{n}\cdot\mathbf{p}-d|}{|\mathbf{n}|}$).
- Scalar triple product = 0 is the coplanarity condition — useful for checking answers.
- AQA may ask you to find a line of intersection of two planes — always use elimination then introduce a parameter.

Summary of Key Vector Formulae

Quantity	Formula
Dot product	$\mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$
Cross product magnitude	$\lvert\mathbf{a}\times\mathbf{b}\rvert = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$
Unit vector	$\hat{\mathbf{a}} = \frac{\mathbf{a}}{\lvert\mathbf{a}\rvert}$
Distance point to line	$\frac{\lvert(\mathbf{p}-\mathbf{a})\times\mathbf{d}\rvert}{\lvert\mathbf{d}\rvert}$
Distance point to plane $\mathbf{n}\cdot\mathbf{r} = d$	$\frac{\lvert\mathbf{n}\cdot\mathbf{p}-d\rvert}{\lvert\mathbf{n}\rvert}$
Distance between skew lines	$\frac{\lvert(\mathbf{c}-\mathbf{a})\cdot(\mathbf{b}\times\mathbf{d})\rvert}{\lvert\mathbf{b}\times\mathbf{d}\rvert}$
Volume of parallelepiped	$\lvert[\mathbf{a},\mathbf{b},\mathbf{c}]\rvert = \lvert\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})\rvert$

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Area of triangle ABC	$\frac{1}{2}\lvert\overrightarrow{AB} \times \overrightarrow{AC}\rvert$
Angle between lines	$\cos\theta = \dfrac{\lvert\mathbf{d}_1 \cdot \mathbf{d}_2\rvert}{\lvert\mathbf{d}_1\rvert\lvert\mathbf{d}_2\rvert}$
Angle between planes	$\cos\theta = \dfrac{\lvert\mathbf{n}_1 \cdot \mathbf{n}_2\rvert}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}$

Position Vectors and Geometry

Example — Proving a quadrilateral is a parallelogram:
$A = (1, 0, 1),\ B = (3, 1, 2),\ C = (4, 3, 3),\ D = (2, 2, 2).$
$\overrightarrow{AB} = (2, 1, 1),\ \overrightarrow{DC} = (2, 1, 1)$. Equal, so $AB \parallel DC$ and $|AB| = |DC|$.
$\overrightarrow{AD} = (1, 2, 1),\ \overrightarrow{BC} = (1, 2, 1)$. Equal. So $ABCD$ is a parallelogram.

Example — Centroid of a triangle:
The centroid G of triangle ABC has position vector $\mathbf{g} = \dfrac{\mathbf{a} + \mathbf{b} + \mathbf{c}}{3}$.
For $A = (2, 4, 0),\ B = (0, 2, 6),\ C = (4, 0, 3)$: $G = \left(\dfrac{6}{3}, \dfrac{6}{3}, \dfrac{9}{3}\right) = (2, 2, 3)$.

Example — Verify four points are coplanar:
$A = (1, 1, 0),\ B = (2, 0, 1),\ C = (0, 2, -1),\ D = (3, 1, 2).$
$\overrightarrow{AB} = (1, -1, 1),\ \overrightarrow{AC} = (-1, 1, -1),\ \overrightarrow{AD} = (2, 0, 2).$
$[\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}] = \overrightarrow{AB} \cdot (\overrightarrow{AC} \times \overrightarrow{AD}).$
$\overrightarrow{AC} \times \overrightarrow{AD} = (1 \cdot 2 - (-1) \cdot 0,\ (-1) \cdot 2 - (-1) \cdot 2,\ (-1) \cdot 0 - 1 \cdot 2) = (2, 0, -2).$
$\overrightarrow{AB} \cdot (2, 0, -2) = 2 + 0 - 2 = 0$. Coplanar. ✓

Lines in 3D — All Forms

A line through point $\mathbf{a}$ in direction $\mathbf{d}$ can be written as:

- **Vector form:** $\mathbf{r} = \mathbf{a} + t\mathbf{d}$
- **Cartesian/symmetric form:** $\dfrac{x - a_1}{d_1} = \dfrac{y - a_2}{d_2} = \dfrac{z - a_3}{d_3} = t$
- **Parametric equations:** $x = a_1 + td_1,\ y = a_2 + td_2,\ z = a_3 + td_3$

Example — Convert to symmetric form:
Line $\mathbf{r} = (2, -1, 3) + t(1, 2, -2)$ becomes:
$\dfrac{x-2}{1} = \dfrac{y+1}{2} = \dfrac{z-3}{-2}$

Example — Find direction from symmetric form:
$\dfrac{x-1}{3} = \dfrac{y-2}{-1} = \dfrac{z}{4}$ gives point $(1, 2, 0)$ and direction $(3, -1, 4)$.

Example — Nearest points on two skew lines:
For $\ell_1 : \mathbf{r} = (1, 0, 0) + s(1, 1, 0)$ and $\ell_2 : \mathbf{r} = (0, 1, 0) + t(0, 0, 1)$:
Connecting vector: $(1 + s, -1 + s, -t)$. Perpendicular to both directions:
$\cdot(1, 1, 0)$: $(1 + s) + (-1 + s) = 0 \Rightarrow 2s = 0 \Rightarrow s = 0$.
$\cdot(0, 0, 1)$: $-t = 0 \Rightarrow t = 0$.
Nearest points: $(1, 0, 0)$ on ℓ_1 and $(0, 1, 0)$ on ℓ_2. Distance $= \sqrt{1 + 1} = \sqrt{2}$.

Projections

Projection of $\mathbf{a}$ onto $\mathbf{b}$:

$$\text{proj}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2}\mathbf{b}, \qquad \text{scalar projection} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$$

The component of $\mathbf{a}$ perpendicular to $\mathbf{b}$: $\mathbf{a} - \text{proj}_{\mathbf{b}}\mathbf{a}$.

Example: Project $\mathbf{a} = (3, 1, 2)$ onto $\mathbf{b} = (1, 0, 1)$.
$\dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} = \dfrac{3 + 0 + 2}{2} = \dfrac{5}{2}$. Projection: $\dfrac{5}{2}(1, 0, 1) = \left(\dfrac{5}{2}, 0, \dfrac{5}{2}\right)$.

Perpendicular component: $(3,1,2) - \left(\frac{5}{2}, 0, \frac{5}{2}\right) = \left(\frac{1}{2}, 1, -\frac{1}{2}\right)$.

Projection tip: The foot of perpendicular from point P to a line is the projection of $\overrightarrow{AP}$ onto the direction $\mathbf{d}$, added back to A. This gives $F = A + \frac{(P-A)\cdot\mathbf{d}}{|\mathbf{d}|^2}\mathbf{d}$. Check: $\overrightarrow{FP}\cdot\mathbf{d} = 0$.

Note: The Gram–Schmidt process uses repeated projection to build an orthogonal basis; this concept underpins many of the distance and perpendicularity calculations in this topic.

Remember: Two vectors are perpendicular iff their dot product is zero. Two lines are parallel iff their direction vectors are scalar multiples. These two facts underpin nearly every vector question.

Cross product direction: Use the right-hand rule — curl fingers from $\mathbf{a}$ to $\mathbf{b}$; thumb points in direction of $\mathbf{a}\times\mathbf{b}$.